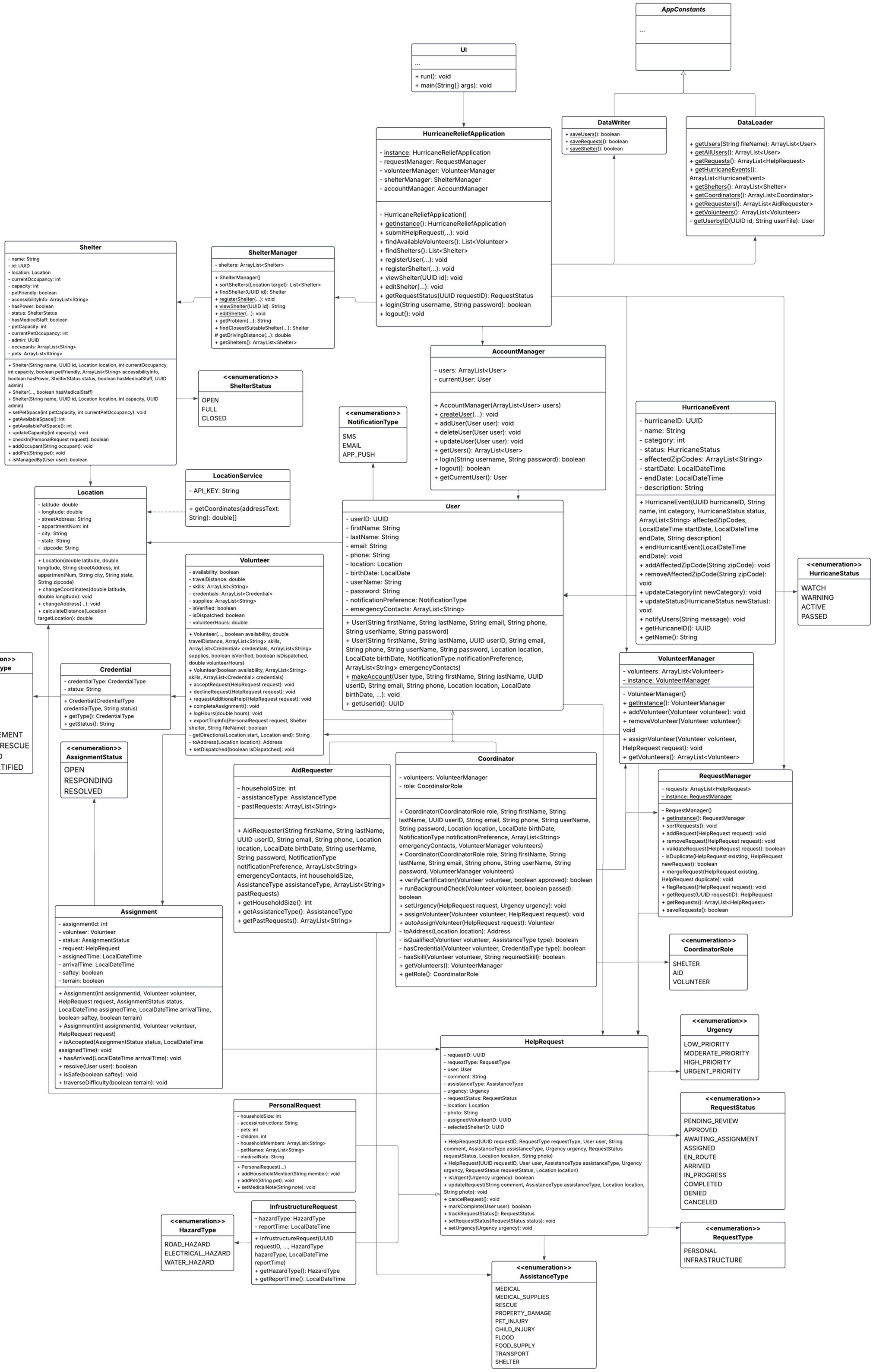

AppConstants
...
UI
...
+ run(): void
+ main(String[] args): void
DataWriter
+ saveUsers(): boolean
+ saveRequests(): boolean
+ saveShelter(): boolean
DataLoader
+ getUsers(String fileName): ArrayList<User>
+ getAllUsers(): ArrayList<User>
+ getRequests(): ArrayList<HelpRequest>
+ getHurricaneEvents(): ArrayList<HurricaneEvent>
+ getShelters(): ArrayList<Shelter>
+ getCoordinators(): ArrayList<Coordinator>
+ getRequesters(): ArrayList<AidRequester>
+ getVolunteers(): ArrayList<Volunteer>
- getUserbyID(UUID id, String userFile): User
HurricaneReliefApplication
- instance: HurricaneReliefApplication
- requestManager: RequestManager
- volunteerManager: VolunteerManager
- shelterManager: ShelterManager
- accountManager: AccountManager
- HurricaneReliefApplication()
+ getInstance(): HurricaneReliefApplication
+ submitHelpRequest(...): void
+ findAvailableVolunteers(): List<Volunteer>
+ findShelters(): List<Shelter>
+ registerUser(...): void
+ registerShelter(...): void
+ viewShelter(UUID id): void
+ editShelter(...): void
+ getRequestStatus(UUID requestID): RequestStatus
+ login(String username, String password): boolean
+ logout(): void
Shelter
- name: String
- id: UUID
- location: Location
- currentOccupancy: int
- capacity: int
- petFriendly: boolean
- accessibilityInfo: ArrayList<String>
- hasPower: boolean
- status: ShelterStatus
- hasMedicalStaff: boolean
- petCapacity: int
- currentPetOccupancy: int
- admin: UUID
- occupants: ArrayList<String>
- pets: ArrayList<String>
+ Shelter(String name, UUID id, Location location, int currentOccupancy, int capacity, boolean petFriendly, ArrayList<String> accessibilityInfo, boolean hasPower, ShelterStatus status, boolean hasMedicalStaff, UUID admin)
+ Shelter(..., boolean hasMedicalStaff)
+ Shelter(String name, UUID id, Location location, int capacity, UUID admin)
+ setPetSpace(int petCapacity, int currentPetOccupancy): void
+ getAvailableSpace(): int
+ getAvailablePetSpace(): int
+ updateCapacity(int capacity): void
+ checkIn(PersonalRequest request): boolean
+ addOccupant(String occupant): void
+ addPet(String pet): void
+ isManagedBy(User user): boolean
ShelterManager
- shelters: ArrayList<Shelter>
+ ShelterManager()
+ sortShelters(Location target): List<Shelter>
+ findShelter(UUID id): Shelter
+ registerShelter(...): void
+ viewShelter(UUID id): String
+ editShelter(...): void
+ getProblem(...): String
+ findClosestSuitableShelter(...): Shelter
getDrivingDistance(...): double
+ getShelters(): ArrayList<Shelter>
<<enumeration>> ShelterStatus
OPEN
FULL
CLOSED
AccountManager
- users: ArrayList<User>
- currentUser: User
+ AccountManager(ArrayList<User> users)
+ createUser(...): void
+ addUser(User user): void
+ deleteUser(User user): void
+ updateUser(User user): void
+ getUsers(): ArrayList<User>
+ login(String username, String password): boolean
+ logout(): boolean
+ getCurrentUser(): User
<<enumeration>> NotificationType
SMS
EMAIL
APP_PUSH
HurricaneEvent
- hurricaneID: UUID
- name: String
- category: int
- status: HurricaneStatus
- affectedZipCodes: ArrayList<String>
- startDate: LocalDateTime
- endDate: LocalDateTime
- description: String
+ HurricaneEvent(UUID hurricaneID, String name, int category, HurricaneStatus status, ArrayList<String> affectedZipCodes, LocalDateTime startDate, LocalDateTime endDate, String description)
+ endHurricantEvent(LocalDateTime endDate): void
+ addAffectedZipCode(String zipCode): void
+ removeAffectedZipCode(String zipCode): void
+ updateCategory(int newCategory): void
+ updateStatus(HurricaneStatus newStatus): void
+ notifyUsers(String message): void
+ getHuricaneID(): UUID
+ getName(): String
<<enumeration>> HurricaneStatus
WATCH
WARNING
ACTIVE
PASSED
Location
- latitude: double
- longitude: double
- streetAddress: String
- appartmentNum: int
- city: String
- state: String
- zipcode: String
+ Location(double latitude, double longitude, String streetAddress, int appartmentNum, String city, String state, String zipcode)
+ changeCoordinates(double latitude, double longitude): void
+ changeAddress(...): void
+ calculateDistance(Location targetLocation): double
LocationService
- API_KEY: String
+ getCoordinates(addressText: String): double[]
User
- userID: UUID
- firstName: String
- lastName: String
- email: String
- phone: String
- location: Location
- birthDate: LocalDate
- userName: String
- password: String
- notificationPreference: NotificationType
- emergencyContacts: ArrayList<String>
+ User(String firstName, String lastName, String email, String phone, String userName, String password)
+ User(String firstName, String lastName, UUID userID, String email, String phone, String userName, String password, Location location, LocalDate birthDate, NotificationType notificationPreference, ArrayList<String> emergencyContacts)
+ makeAccount(User type, String firstName, String lastName, UUID userID, String email, String phone, Location location, LocalDate birthDate, ...): void
+ getUserId(): UUID
Volunteer
- availability: boolean
- travelDistance: double
- skills: ArrayList<String>
- credentials: ArrayList<Credential>
- supplies: ArrayList<String>
- isVerified: boolean
- isDispatched: boolean
- volunteerHours: double
+ Volunteer(..., boolean availability, double travelDistance, ArrayList<String> skills, ArrayList<Credential> credentials, ArrayList<String> supplies, boolean isVerified, boolean isDispatched, double volunteerHours)
+ Volunteer(boolean availability, ArrayList<String> skills, ArrayList<Credential> credentials)
+ acceptRequest(HelpRequest request): void
+ declineRequest(HelpRequest request): void
+ requestAdditionalHelp(HelpRequest request): void
+ completeAssignment(): void
+ logHours(double hours): void
+ exportTripInfo(PersonalRequest request, Shelter shelter, String fileName): boolean
- getDirections(Location start, Location end): String
- toAddress(Location location): Address
+ setDispatched(boolean isDispatched): void
VolunteerManager
- volunteers: ArrayList<Volunteer>
- instance: VolunteerManager
- VolunteerManager()
+ getInstance(): VolunteerManager
+ addVolunteer(Volunteer volunteer): void
+ removeVolunteer(Volunteer volunteer): void
+ assignVolunteer(Volunteer volunteer, HelpRequest request): void
+ getVolunteers(): ArrayList<Volunteer>
<<enumeration>> CredentialType
EMT
PARAMEDIC
NURSE
DOCTOR
FIREFIGHTER
LAW_ENFORCEMENT
SEARCH_AND_RESCUE
CPR_CERTIFIED
FIRST_AID_CERTIFIED
Credential
- credentialType: CredentialType
- status: String
+ Credential(CredentialType credentialType, String status)
+ getType(): CredentialType
+ getStatus(): String
<<enumeration>> AssignmentStatus
OPEN
RESPONDING
RESOLVED
AidRequester
- householdSize: int
- assistanceType: AssistanceType
- pastRequests: ArrayList<String>
+ AidRequester(String firstName, String lastName, UUID userID, String email, String phone, Location location, LocalDate birthDate, String userName, String password, NotificationType notificationPreference, ArrayList<String> emergencyContacts, int householdSize, AssistanceType assistanceType, ArrayList<String> pastRequests)
+ getHouseholdSize(): int
+ getAssistanceType(): AssistanceType
+ getPastRequests(): ArrayList<String>
Coordinator
- volunteers: VolunteerManager
- role: CoordinatorRole
+ Coordinator(CoordinatorRole role, String firstName, String lastName, UUID userID, String email, String phone, String userName, String password, Location location, LocalDate birthDate, NotificationType notificationPreference, ArrayList<String> emergencyContacts, VolunteerManager volunteers)
+ Coordinator(CoordinatorRole role, String firstName, String lastName, String email, String phone, String userName, String password, VolunteerManager volunteers)
+ verifyCertification(Volunteer volunteer, boolean approved): boolean
+ runBackgroundCheck(Volunteer volunteer, boolean passed): boolean
+ setUrgency(HelpRequest request, Urgency urgency): void
+ assignVolunteer(Volunteer volunteer, HelpRequest request): void
+ autoAssignVolunteer(HelpRequest request): Volunteer
- toAddress(Location location): Address
- isQualified(Volunteer volunteer, AssistanceType type): boolean
- hasCredential(Volunteer volunteer, CredentialType type): boolean
- hasSkill(Volunteer volunteer, String requiredSkill): boolean
+ getVolunteers(): VolunteerManager
+ getRole(): CoordinatorRole
RequestManager
- requests: ArrayList<HelpRequest>
- instance: RequestManager
- RequestManager()
+ getInstance(): RequestManager
+ sortRequests(): void
+ addRequest(HelpRequest request): void
+ removeRequest(HelpRequest request): void
+ validateRequest(HelpRequest request): boolean
- isDuplicate(HelpRequest existing, HelpRequest newRequest): boolean
+ mergeRequest(HelpRequest existing, HelpRequest duplicate): void
+ flagRequest(HelpRequest request): void
+ getRequest(UUID requestID): HelpRequest
+ getRequests(): ArrayList<HelpRequest>
+ saveRequests(): boolean
<<enumeration>> CoordinatorRole
SHELTER
AID
VOLUNTEER
Assignment
- assignmentId: int
- volunteer: Volunteer
- status: AssignmentStatus
- request: HelpRequest
- assignedTime: LocalDateTime
- arrivalTime: LocalDateTime
- saftey: boolean
- terrain: boolean
+ Assignment(int assignmentId, Volunteer volunteer, HelpRequest request, AssignmentStatus status, LocalDateTime assignedTime, LocalDateTime arrivalTime, boolean saftey, boolean terrain)
+ Assignment(int assignmentId, Volunteer volunteer, HelpRequest request)
+ isAccepted(AssignmentStatus status, LocalDateTime assignedTime): void
+ hasArrived(LocalDateTime arrivalTime): void
+ resolve(User user): boolean
+ isSafe(boolean saftey): void
+ traverseDifficulty(boolean terrain): void
HelpRequest
- requestID: UUID
- requestType: RequestType
- user: User
- comment: String
- assistanceType: AssistanceType
- urgency: Urgency
- requestStatus: RequestStatus
- location: Location
- photo: String
- assignedVolunteerID: UUID
- selectedShelterID: UUID
+ HelpRequest(UUID requestID, RequestType requestType, User user, String comment, AssistanceType assistanceType, Urgency urgency, RequestStatus requestStatus, Location location, String photo)
+ HelpRequest(UUID requestID, User user, AssistanceType assistanceType, Urgency urgency, RequestStatus requestStatus, Location location)
+ isUrgent(Urgency urgency): boolean
+ updateRequest(String comment, AssistanceType assistanceType, Location location, String photo): void
+ cancelRequest(): void
+ markComplete(User user): boolean
+ trackRequestStatus(): RequestStatus
+ setRequestStatus(RequestStatus status): void
+ setUrgency(Urgency urgency): void
<<enumeration>> Urgency
LOW_PRIORITY
MODERATE_PRIORITY
HIGH_PRIORITY
URGENT_PRIORITY
<<enumeration>> RequestStatus
PENDING_REVIEW
APPROVED
AWAITING_ASSIGNMENT
ASSIGNED
EN_ROUTE
ARRIVED
IN_PROGRESS
COMPLETED
DENIED
CANCELED
<<enumeration>> RequestType
PERSONAL
INFRASTRUCTURE
PersonalRequest
- householdSize: int
- accessInstructions: String
- pets: int
- children: int
- householdMembers: ArrayList<String>
- petNames: ArrayList<String>
- medicalNote: String
+ PersonalRequest(...)
+ addHouseholdMember(String member): void
+ addPet(String pet): void
+ setMedicalNote(String note): void
InfrustructureRequest
- hazardType: HazardType
- reportTime: LocalDateTime
+ InfrustructureRequest(UUID requestID, ..., HazardType hazardType, LocalDateTime reportTime)
+ getHazardType(): HazardType
+ getReportTime(): LocalDateTime
<<enumeration>> HazardType
ROAD_HAZARD
ELECTRICAL_HAZARD
WATER_HAZARD
<<enumeration>> AssistanceType
MEDICAL
MEDICAL_SUPPLIES
RESCUE
PROPERTY_DAMAGE
PET_INJURY
CHILD_INJURY
FLOOD
FOOD_SUPPLY
TRANSPORT
SHELTER